Accelerator Physics Course

Emittance Measurements at the ALBA Linac

Instructor: Laura Torino
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Repository: GitHub-Repo

1. Background

Particle accelerators require accurate beam diagnostics to determine and optimize beam quality and dynamics. Using appropriate instrumentation and measurement techniques, the particle beam inside an accelerator can be characterized by obtaining information about beam parameters such as position, intensity, and size. In this experiment, beam size is measured and analyzed at the ALBA Linac by conducting a quadrupole scan around the beam waist. The beam transport through the transfer line is represented in Figure 1 and machine parameters are introduced in Table 1.

To describe beam transport through the quadrupole section and determine the beam-size evolution as a function of quadrupole current, the following calibrated relations are used:

$$E = A' I_{bend} + A'_0, \quad \text{[MeV]}, \quad \text{dipole energy calibration,} \tag{1}$$

$$K_1 = \frac{I_{quad} F}{E}, \quad [\mathrm{m}^{-2}], \quad \text{quadrupole strength,} \tag{2}$$

$$k = K_1 L_Q, \quad [\mathrm{m}^{-1}], \quad \text{integrated quadrupole strength,} \tag{3}$$

$$s_k = \mathrm{sign}(K_1)\sqrt{|K_1|}, \quad [\mathrm{m}^{-1}], \quad \text{quadrupole strength square root,} \tag{4}$$

$$Q_{21} = s_k \sin\left(|s_k L_Q|\right), \quad [\mathrm{m}^{-1}], \quad \text{quadrupole focal strength.} \tag{5}$$

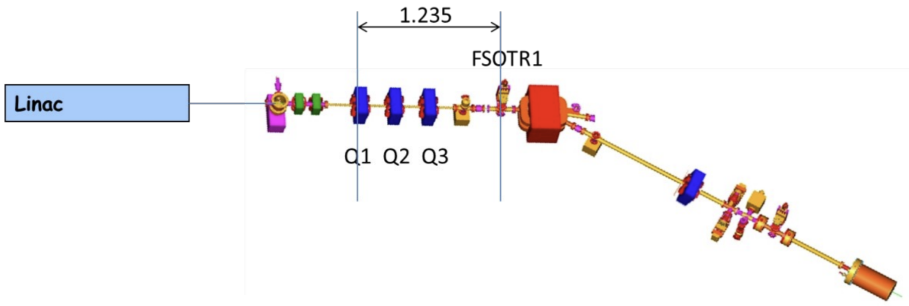


Figure 1: Sketch of the Linac, transfer line, and diagnostics line.

Parameter	Symbol	Value	Units
Beam energy	E	~ 80	MeV
Dipole current	I_{bend}	~ 110	A
Dipole calibration coefficient	A'	0.7058	MeV/A
Dipole offset	A'_0	0.2448	MeV
Quadrupole current	I_{quad}	$[-5, +5]$	A
Distance Q1 $\rightarrow$ FSOTR	d	1.235	m
Quadrupole length	L_Q	0.126	m
Quadrupole calibration factor	F	387.65	MeV/A/m^2

Table 1: Machine parameters and calibration constants.

Multi-Bunch Mode

2. Beam Characteristics

The longitudinal characteristics of the beam can be determined using the Linac Beam Charge Monitor (BCM), shown in Figure 7, and the Fast Current Transformer. The measurements are

Beam charge	$0.5nC$
Bunch train length	$78ns$
Number of bunches	39

The dipole magnet intensity is measured as $I_{bend} = 113.3\,\text{A}$. Then, using the dipole calibration relation in Eq. 1, the beam energy is $E = 80.21MeV$, which is in agreement with the expected result of $\approx 80MeV$ and with the measurement of $80.196MeV$ shown in the GUI presented in Figure 8.

3. Horizontal Emittance Measurement

The effects of quadrupole strength on the horizontal beam size can be studied with an intensity scan of quadrupole Q1 around the beam waist. The quadrupole intensity I_{quad} and the horizontal RMS beam size σ_x are measured with the diagnostics monitor and presented in Table 2. Applying these measurements to the quadrupole calibration relations, the quadrupole focal strength (Eq. 5) and the integrated strength (Eq. 3) are determined.

Additionally, the validity of the thin lens approximation for a quadrupole can be assessed by comparing $K_1 L_Q$, the integrated strength that represents the thin lens approximation, with Q_{21}, the exact expression derived from the quadrupole transfer matrix.

$$\%\text{diff thin lens} = \frac{(K_1 L_Q - Q_{21})}{Q_{21}} \times 100 \tag{6}$$

The thin lens approximation assumes $Q_{21} = k = K_1 L_Q$, which holds when $|s_k L_Q| \ll 1$ so that $\sin(|s_k L_Q|) \approx |s_k L_Q|$. As shown in Table 2, the percentage difference ranges from $(0.26 \pm 0.13)\%$ at the lowest current setting to $(2.87 \pm 0.13)\%$ at the highest, remaining below 3% across the full scan range. These deviations are within the measurement uncertainties on σ^2, confirming that the thin lens approximation $Q_{21} \approx K_1 L_Q$ holds to a good degree for the quadrupole strengths and characteristics ($L_Q = 0.126\,\text{m}$) used in this experiment. The deviation grows with current because the argument $|s_k L_Q|$ increases, making the small-angle approximation $\sin(x) \approx x$ less accurate.

The data calculated in Table 2 is used to obtain the fit presented in Figure 2b to the parabola

$$\sigma^2 = A\,(Q_{21} - B)^2 + C \tag{7}$$

where B corresponds to the waist focal strength ($Q_{21}^{\text{waist}} = 0.550\,\text{m}^{-1}$) marked in gray, $C = \sigma^2_{\text{min}} = 0.110\,\text{mm}^2$ is the minimum beam size squared at the waist marked in green, and A

controls the beam divergence. From the fit parameters, the geometric emittance is obtained as

$$\varepsilon_x = \frac{\sqrt{A \cdot C}}{d^2} = (0.094 \pm 0.006)\,\mu\mathrm{m} \cdot \mathrm{rad} \tag{8}$$

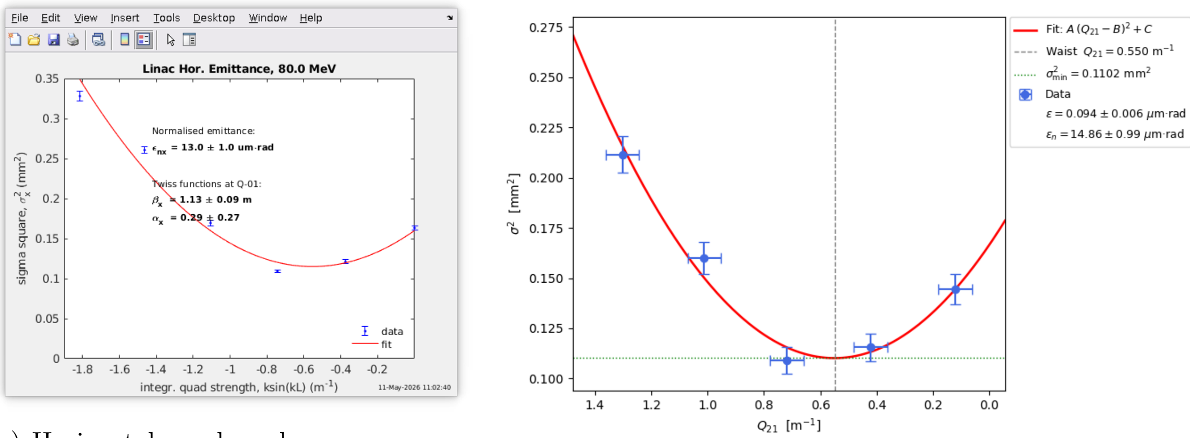
where $d = 1.235\,\mathrm{m}$ is the drift length from Q1 to FSOTR1. From this, the normalized emittance is related to the geometric emittance by

$$\varepsilon_n x = \beta\gamma\,\varepsilon = (14.86 \pm 0.99)\,\mu\mathrm{m} \cdot \mathrm{rad} \tag{9}$$

This result is compatible with the ALBA control room GUI (Figure 2a), which reports $\varepsilon_{nx}^{GUI} = (13.0 \pm 1.0)\,\mu\mathrm{m} \cdot \mathrm{rad}$, the small discrepancy being attributable to the limited precision with which data was recorded manually from the control room GUI.

Table 2: Quad scan around the waist current for the horizontal beam size in multi-bunch mode.

	I_{quads} (A)	Q_{21} (m^{-1})	$K_1 L_Q$ (m^{-1})	σ (mm)	σ^2 (mm^2)	% diff thin lens
$I_w + 2\Delta I$	2.20 ± 0.10	1.3 ± 0.1	1.34 ± 0.06	0.46 ± 0.01	0.212 ± 0.009	2.87 ± 0.13
$I_w + \Delta I$	1.70 ± 0.10	1.0 ± 0.1	1.04 ± 0.06	0.40 ± 0.01	0.160 ± 0.008	2.21 ± 0.13
I_w	1.20 ± 0.10	0.7 ± 0.1	0.73 ± 0.06	0.33 ± 0.01	0.109 ± 0.007	1.55 ± 0.13
$I_w - \Delta I$	0.70 ± 0.10	0.4 ± 0.1	0.43 ± 0.06	0.34 ± 0.01	0.116 ± 0.007	0.90 ± 0.13
$I_w - 2\Delta I$	0.20 ± 0.10	0.1 ± 0.1	0.12 ± 0.06	0.38 ± 0.01	0.144 ± 0.008	0.26 ± 0.13



(a) Horizontal quadrupole scan measurements from the ALBA control room GUI.

(b) Horizontal quadrupole scan fit from Table 2 data.

Figure 2: Horizontal emittance measurement in multi-bunch mode.

4. Vertical Emittance Measurement

Following the same procedure described in Section 3, the scan around the waist is repeated for the vertical plane. The measured data and calculated quantities are presented in Table 3 and the corresponding parabolic fit is shown in Figure 3. The first notable observation from the measured data is that the quadrupole currents around the waist are now negative, whereas in the horizontal case they were positive. This is consistent with quadrupoles: the same magnet that focuses horizontally for positive currents acts as a focusing element in the vertical plane when the current

is reversed, as the magnetic field gradient changes sign. Consequently, the Q_{21} and K_1L_Q values in Table 3 are also negative. Regarding the validity of the thin lens approximation in the vertical case, the percentage differences again do not exceed 3%, as in the horizontal case. Therefore, it is also a good approximation for values near the waist, where the current is sufficiently small for the $\sin(|s_kL_Q|) \approx |s_kL_Q|$ approximation to be valid and thus $k = K_1L_Q \approx Q_{21}$.
The vertical measurements can again be plotted to fit the parabola in Eq. 7, obtaining the results shown in Fig. 3b. The obtained geometric and normalized emittances are

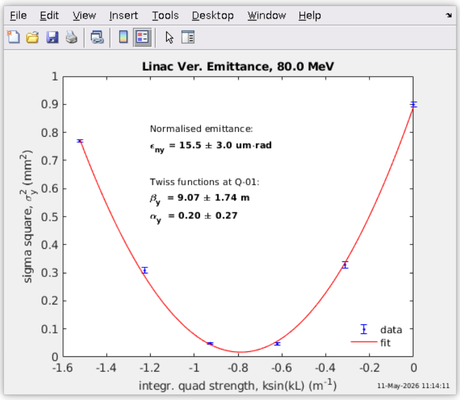
$$\varepsilon_y = (0.077 \pm 0.008)\,\mu\text{m} \cdot \text{rad}\varepsilon_n y = (12.15 \pm 1.24)\,\mu\text{m} \cdot \text{rad}$$

Then it is found that the normalized vertical emittance is sufficiently similar to the horizontal emittance to a degree compatible with an approximately symmetric beam. Additionally, this fit closely corresponds with the one performed by the ALBA GUI shown in Figure 3a, which shows a normalized emittance of $\varepsilon_{ny}^{GUI} = (15.5 \pm 3)\,\mu\text{m} \cdot \text{rad}$, compatible within the uncertainties.
A notable difference with respect to the horizontal case is the significantly smaller minimum beam size at the waist: $\sigma^2_{\text{min}} = 0.009\,\text{mm}^2$ in the vertical plane compared to $\sigma^2_{\text{min}} = 0.110\,\text{mm}^2$ in the horizontal plane. This is explained by the beta functions at the quadrupole. Using the relations $\varepsilon = \sqrt{AC}/d^2$ and $\beta_0 = \sqrt{A/C}$ given by the fit parameters, it can be shown that $C = \sigma^2_{\text{min}} = \varepsilon d^2/\beta_0$. Therefore, a larger beta function at Q1 directly implies a smaller minimum beam size at FSOTR. In this case, the Twiss parameters reported by the ALBA control room GUI at Q1 are $\beta_x = 1.13\,\text{m}$ and $\beta_y = 9.07\,\text{m}$; therefore, the vertical beta function is approximately eight times larger than the horizontal one. As a consequence, Q1 is able to produce a smaller vertical beam size, which reflects the preferred optics configuration for this quadrupole in normal injection operation at ALBA.

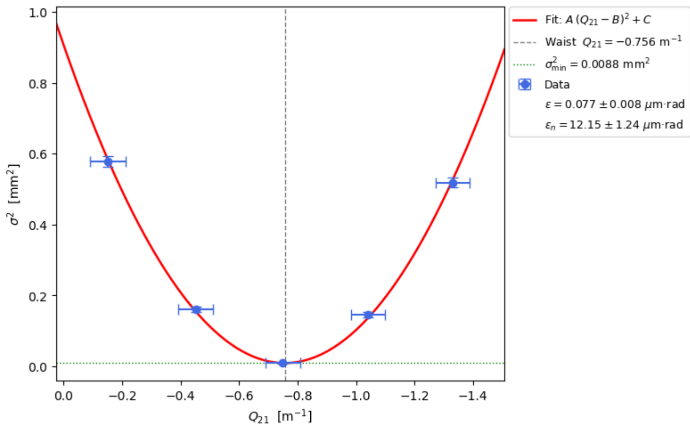
Using the relations $\varepsilon = \sqrt{AC}/d^2$ and $\beta_0 = \sqrt{A/C}$ given by the fit parameters, one can show that $C = \sigma^2_{\text{min}} = \varepsilon d^2/\beta_0$. Therefore, a larger beta function at Q1 directly implies a smaller minimum beam size at FSOTR.

Table 3: Quad scan around the waist current for the vertical beam size in multi-bunch mode.

	I_{quads} (A)	Q_{21} (m^{-1})	K_1L_Q (m^{-1})	σ (mm)	σ^2 (mm^2)	% diff thin lens
$I_w + 2\Delta I$	-0.25 ± 0.10	-0.2 ± 0.1	-0.15 ± 0.06	0.76 ± 0.01	0.578 ± 0.015	0.32 ± 0.13
$I_w + \Delta I$	-0.75 ± 0.10	-0.5 ± 0.1	-0.46 ± 0.06	0.40 ± 0.01	0.160 ± 0.008	0.97 ± 0.13
I_w	-1.25 ± 0.10	-0.7 ± 0.1	-0.76 ± 0.06	0.09 ± 0.01	0.008 ± 0.002	1.62 ± 0.13
$I_w - \Delta I$	-1.75 ± 0.10	-1.0 ± 0.1	-1.07 ± 0.06	0.38 ± 0.01	0.144 ± 0.008	2.27 ± 0.13
$I_w - 2\Delta I$	-2.25 ± 0.10	-1.3 ± 0.1	-1.37 ± 0.06	0.72 ± 0.01	0.518 ± 0.014	2.94 ± 0.13



(a) Vertical quadrupole scan measurements from the ALBA control room GUI.



(b) Vertical quadrupole scan fit from Table 3 data.

Figure 3: Vertical emittance measurement in multi-bunch mode.

Single-Bunch Mode

5. Beam Characteristics

Following the same Linac settings as in the multi-bunch mode and with the same beam energy of $\approx 80MeV$, the characteristics of the beam in single-bunch mode are

Beam charge	$0.04nC$
Bunch train length	$2ns$
Number of bunches	1

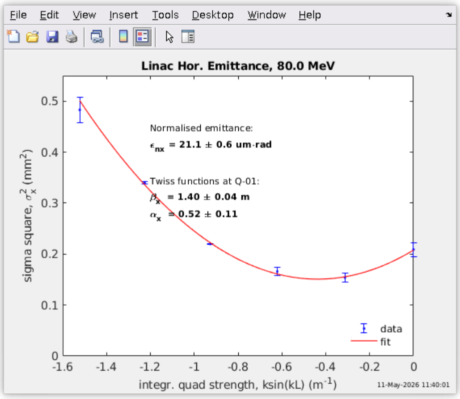
6. Horizontal Emittance Measurement

The experimental results in the horizontal case for single-bunch mode are shown in Table 4, and the fit is represented in Figure 4. Similar observations to the multi-bunch case can be made regarding the validity of the thin lens approximation and beam size. As for the quadrupole current, the waist is displaced to $I_{quad} = 0.75A$ compared with $I_{quad} = 1.2A$ in the multi-bunch case, suggesting a slight change in the horizontal beam optics.

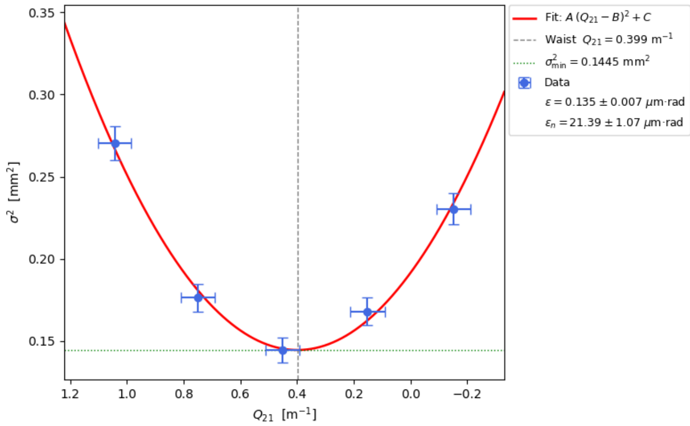
Notably, the normalized emittance increases to $\varepsilon_{nx} = (21.4 \pm 1.1)\,\mu\text{m} \cdot \text{rad}$, compatible with the ALBA GUI value of $\varepsilon_{nx}^{GUI} = (21.1 \pm 0.6)\,\mu\text{m} \cdot \text{rad}$. This discrepancy can be attributed to the charge of a single bunch. In the multi-bunch mode, the total $0.5nC$ charge was distributed among 39 bunches, which represents a $0.013nC$ charge per bunch, whereas in the single-bunch case, the total $0.04nC$ charge is concentrated in a single bunch. Given this higher charge density, the electrons inside the same bunch repel each other more, giving rise to a higher emittance. Therefore, the increase in the space-charge force in the single-bunch case explains the higher normalized emittance found in this experiment.

Table 4: Quad scan around the waist current for the horizontal beam size in single-bunch mode.

	I_{quads} (A)	Q_{21} (m^{-1})	K_1L_Q (m^{-1})	σ (mm)	σ^2 (mm^2)	% diff thin lens
$I_w + 2\Delta I$	1.75 ± 0.10	1.0 ± 0.1	1.07 ± 0.06	0.52 ± 0.01	0.270 ± 0.010	2.27 ± 0.13
$I_w + \Delta I$	1.25 ± 0.10	0.7 ± 0.1	0.76 ± 0.06	0.42 ± 0.01	0.176 ± 0.008	1.62 ± 0.13
I_w	0.75 ± 0.10	0.5 ± 0.1	0.46 ± 0.06	0.38 ± 0.01	0.144 ± 0.008	0.97 ± 0.13
$I_w - \Delta I$	0.25 ± 0.10	0.2 ± 0.1	0.15 ± 0.06	0.41 ± 0.01	0.168 ± 0.008	0.32 ± 0.13
$I_w - 2\Delta I$	-0.25 ± 0.10	-0.2 ± 0.1	-0.15 ± 0.06	0.48 ± 0.01	0.230 ± 0.010	0.32 ± 0.13



(a) Horizontal quadrupole scan measurements from the ALBA control room GUI.



(b) Horizontal quadrupole scan fit from Table 4 data.

Figure 4: Horizontal emittance measurement in single-bunch mode.

7. Vertical Emittance Measurement

Following the same procedure as in the previous cases, the quadrupole scan results are presented in Table 5 and the fit is shown in Figure 5. The waist current is found at $I_w = -1.25\,\mathrm{A}$, identical to the multi-bunch vertical case, consistent with the vertical optics of the transfer line being unchanged between modes. The fit yields $C = \sigma^2_{\min} = (2.7 \cdot 10^{-04} \pm 1.6 \cdot 10^{-3})\,\mathrm{mm}^2$, where the uncertainty is six times larger than the central value. As a consequence, the geometric emittance $\varepsilon_y = (0.021 \pm 0.063)\,\mu\mathrm{m} \cdot \mathrm{rad}$ and the normalized emittance $\varepsilon_n y = (3.36 \pm 10.03)\,\mu\mathrm{m} \cdot \mathrm{rad}$ carry an uncertainty three times larger than the central value itself and cannot be considered a reliable measurement. Nevertheless, the results suggest that in single-bunch mode the vertical emittance is also lower than the horizontal case. A more appropriate diagnostic for this case would require more precise measurements or a different measurement technique.

Table 5: Quad scan around the waist current for the vertical beam size in single-bunch mode.

	I_{quads} (A)	Q_{21} (m^{-1})	$K_1 L_Q$ (m^{-1})	σ (mm)	σ^2 (mm^2)	% diff thin lens
$I_w + 2\Delta I$	-0.25 ± 0.10	-0.2 ± 0.1	-0.15 ± 0.06	1.10 ± 0.01	1.210 ± 0.022	0.32 ± 0.13
$I_w + \Delta I$	-0.75 ± 0.10	-0.5 ± 0.1	-0.46 ± 0.06	0.51 ± 0.01	0.260 ± 0.010	0.97 ± 0.13
I_w	-1.25 ± 0.10	-0.7 ± 0.1	-0.76 ± 0.06	0.08 ± 0.01	0.006 ± 0.002	1.62 ± 0.13
$I_w - \Delta I$	-1.75 ± 0.10	-1.0 ± 0.1	-1.07 ± 0.06	0.46 ± 0.01	0.212 ± 0.009	2.27 ± 0.13
$I_w - 2\Delta I$	-2.25 ± 0.10	-1.3 ± 0.1	-1.37 ± 0.06	1.40 ± 0.01	1.960 ± 0.028	2.94 ± 0.13

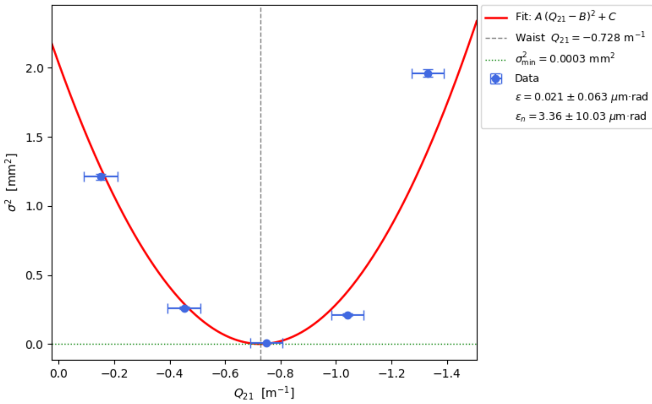


Figure 5: Vertical quadrupole scan fit from Table 5 data.

8. Conclusion

This experiment has characterised the transverse emittance of the ALBA linac beam in both multi-bunch and single-bunch operation modes through quadrupole scan measurements using the FSOTR diagnostic screen. In both modes the thin lens approximation was found to be valid for the scan range, and the results are in agreement with the ALBA control room Linac diagnostics tools. However, the higher charge density per bunch in the single-bunch mode generates a notably larger horizontal emittance. The measured emittances are of the order of 10–20 μm · rad, which is typical for a thermionic-gun linac operating at this energy range.

Appendices

A. Supplementary Control-System Records

This appendix collects the control-room screenshots used to document the beam conditions during the multi-bunch measurements.

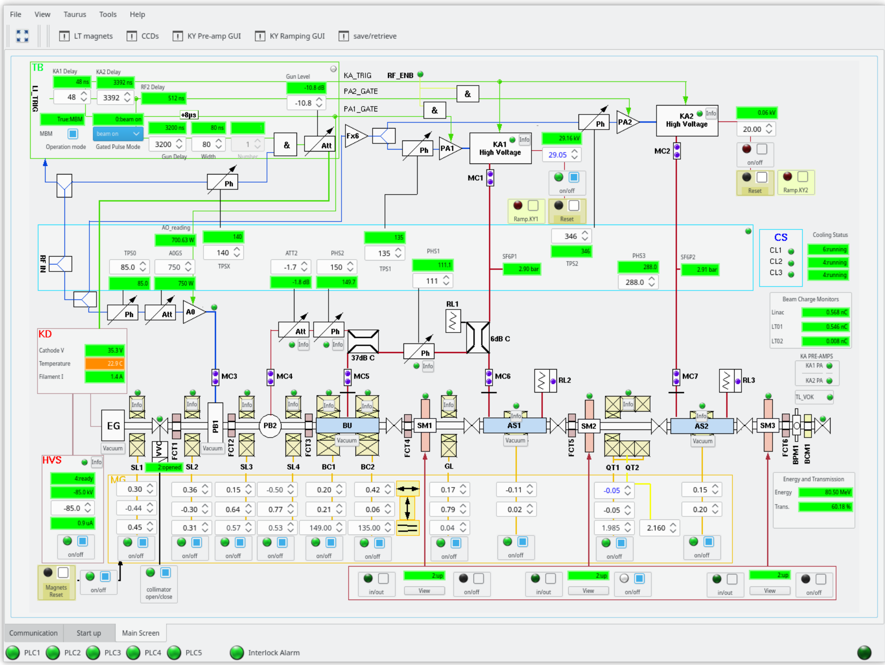


Figure 6: ALBA Linac control-system overview during the multi-bunch measurement. The beam charge monitor readbacks are displayed in the center-right panel.

Figure 7: Beam current monitor signal used to characterize the multi-bunch time structure.

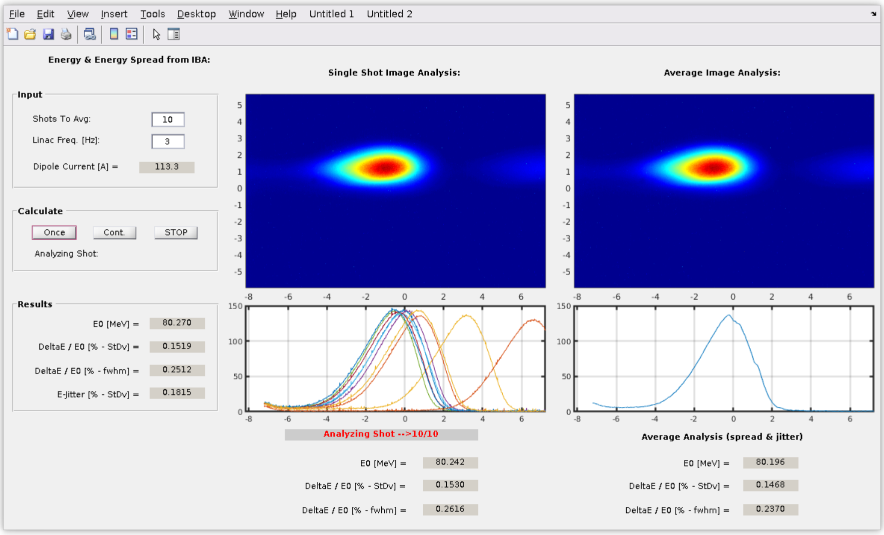


Figure 8: Beam energy analysis for the multi-bunch operating point.